

AKROMID®

A3 GF 30 1 black (2385)

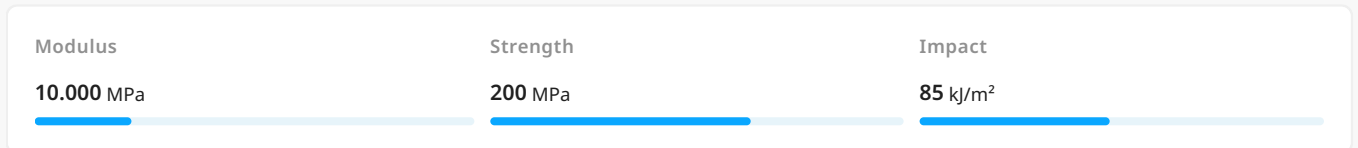
PA66 GF30

AKROMID® A3 GF 30 1 black (2385) is a 30% glass fiber reinforced polyamide 6.6. It is characterised by high stiffness and strength. Furthermore, the material is heat stabilised and therefore perfectly suitable for technical parts in industrial engineering and in the automotive industry.

Features

heat stabilised 130

Properties



Mechanical Properties

Tensile modulus ISO 527-2	1 mm/min d.a.m.	10000 MPa
	1 mm/min conditioned	7100 MPa
Tensile stress at break ISO 527-2	5 mm/min d.a.m.	200 MPa
	5 mm/min conditioned	130 MPa
Tensile strain at break ISO 527-2	5 mm/min d.a.m.	3 %
	5 mm/min conditioned	>= 6 %
Flexural modulus ISO 178	2 mm/min d.a.m.	8800 MPa
	2 mm/min conditioned	7200 MPa
Flexural strength ISO 178	2 mm/min d.a.m.	285 MPa
	2 mm/min conditioned	220 MPa
Charpy impact strength ISO 179-1/1eU	23°C d.a.m.	85 kJ/m²
	23°C conditioned	95 kJ/m²
	-30°C d.a.m.	80 kJ/m²
Charpy notched impact strength ISO 179-1/1eA	23°C d.a.m.	10 kJ/m²
	23°C conditioned	13 kJ/m²
	-30°C d.a.m.	9 kJ/m²

Disclaimer

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Ball indentation hardness

ISO 2039-1

961N/30s | d.a.m.

240 MPa**Thermal Properties****Temperature of deflection under load HDT/A**

ISO 75

1,8 MPa

255 °C**Temperature of deflection under load HDT/B**

ISO 75

0,45 MPa

260 °C**Temperature of deflection under load HDT/C**

ISO 75

8 MPa

210 °C**Melting temperature**

ISO 11357-3

DSC, 10K/min

262 °C**Coefficient of linear thermal expansion**

ISO 11359-1/2

23°C to 80°C | parallel

0,19 10⁻⁴/K

23°C to 80°C | transverse

0,95 10⁻⁴/K**Temperature index for 50% loss of tensile strength**

IEC 60216

5.000 h

150 °C**Flammability****Burning rate**

UL 94

0,8 mm Wall thickness

HB Class**GWFI**

IEC 60695-2-12

1,6 mm Wall thickness

> 650 °C**Burning rate (<100 mm/min)**

FMVSS 302

> 1 mm Thickness

+**General properties****Density**

ISO 1183

23°C

1,36 g/cm³**Humidity absorption**

ISO 1110

70°C, 62% r.H.

1,9 - 2,1 %**Disclaimer**

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Water absorption	23°C, saturated	5,2 - 5,8 %
ISO 62		
Molding shrinkage	flow	0,1 - 0,3 %
ISO 294-4	transverse	0,7 - 0,9 %

Electrical Properties

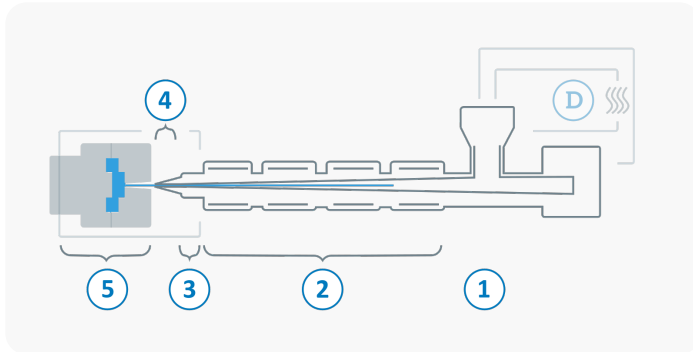
Volume resistivity IEC	d.a.m.	10¹³ Ω x cm
DIN EN IEC 62631-3-2	conditioned	10¹⁰ Ω x cm
Surface resistivity	d.a.m.	10¹² Ω
DIN EN 62631-3-2	conditioned	10¹⁰ Ω
Comparative tracking index	Test liquid A	600 V
IEC 60112		

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Processing

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only de-humidifying or vacuum dryers. Extensive drying can cause filling problem and surface defects.

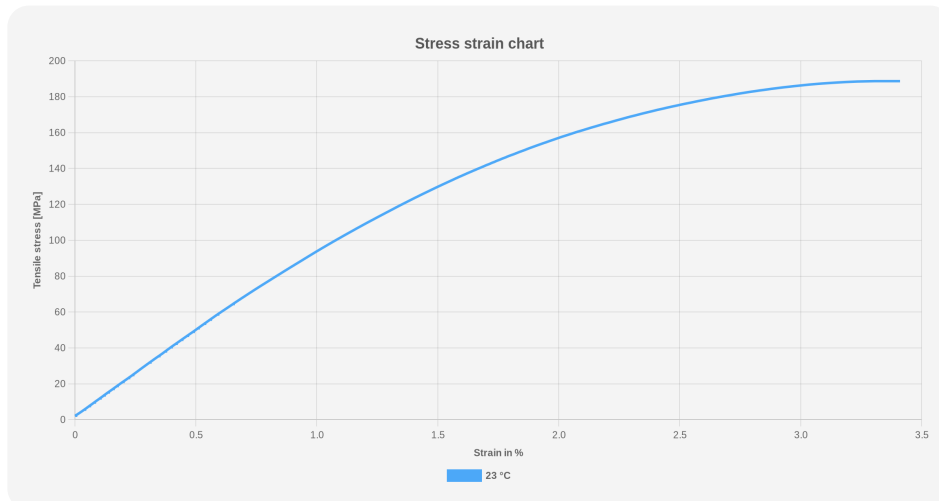
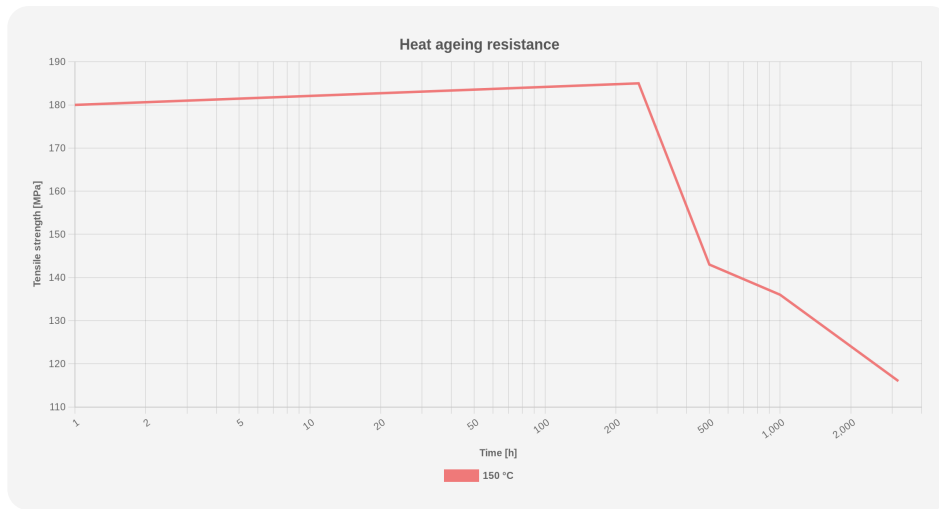


D	Drying time	0 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80 °C
	Processing moisture	0,02 - 0,1 %
1	Feed section	60 - 80 °C
2	Temperature Zone 1 - Zone 4	260 - 300 °C
3	Nozzle temperature	270 - 310 °C
4	Melt temperature	280 - 300 °C
5	Mold temperature	80 - 100 °C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

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Diagrams



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